

The University of Hong Kong

Department of Data and Systems Engineering

M.Sc. (Eng.) in Industrial Engineering and Logistics Management

IMSE7099 Dissertation

Interim Report

One Size Does Not Fit All:

Personal Match and Marketing Message Effectiveness on Social Networks

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Date of submission: 24 February 2026

Abstract

In the era of social media, people's daily lives are almost inseparable from using social media. Besides its central role of connecting people, it has also expanded to other functions, particularly marketing. It is common to see marketing messages posted on social media, as it provides a platform that businesses can use to spread their messages widely and effortlessly through the online network, presenting a new way to reach their target clientele efficiently. However, the efficiency of the messages may significantly vary across audiences. Therefore, it is worthwhile to investigate how the fit between marketing content and individual influencers in social networks influences the overall diffusion and engagement of marketing messages.

This study aims to investigate the role of personal matching between message content and the social network characteristics of the marketing agent in predicting the effectiveness of marketing messages. Using the "WeiMai" social marketing system dataset, which records real-world article sharing and reading behaviors in the interior design industries, to understand how content, marketing agents' network, and their interaction jointly determine the diffusion outcomes, such as total reads and diffusion depth.

The analytical framework consists of three interconnected components, including an article model that captures the semantic and thematic features of the title and title-content consisting of marketing articles which appear in online network sharing, a network model that represents the relational and structural properties of marketing agents within the sharing network, and an article-agent matching model that combines the two to forecast the spreading effect. The three models will be compared with one another. The performance of the article-agent matching model

is the primary interest of this study, which presents the effectiveness of personal matching between messages and networks in social marketing.

The outcomes of this study will include a thorough demonstration of how content-network alignment affects marketing diffusion in social media networks. Data-driven regression models will be developed to predict and optimize marketing performance. These research results should provide valuable suggestions for designing more effective and personalized social marketing strategies, as well as recommendation systems in social marketing. To date, we have successfully integrated heterogeneous data sources by linking marketing agent human-resources profiles with social media identifiers, resulting in a verified user base. A key technical milestone is the deployment of a multi-modal large language model (GPT-4o) pipeline that automatically clusters unstructured marketing articles into semantically coherent topics. In addition, we constructed three interrelated diffusion network datasets at the content, marketing agents' network, and marketing agent–topic levels, and embedded key cascade and diffusion-network metrics within each dataset to support the proposed three-level analytical framework.

Preliminary descriptive analyses indicate that alignment between agent professional background and content characteristics is associated with deeper diffusion cascades. The primary challenges ahead include: establishing a theoretically and empirically defensible definition of the dependent variable while preventing outcome-derived information leakage in predictors; distinguishing variables that genuinely capture diffusion structure from post-hoc or non-diffusion attributes; and developing a robust modeling and validation pipeline capable of fairly comparing content-only, network-only, and integrated approaches.

1. Introduction

1.1 Background of Study

The tremendous growth of social media such as WeChat, Facebook, and Instagram plays a significant role in driving an innovative shift from traditional marketing to digital marketing (Rajendra Kumar et al., 2025). Social media has fundamentally transformed the way businesses disseminate marketing messages and interact with target customers, enabling real-time engagement and large-scale information propagation through online networks.

In online social marketing, a company disseminates promotional content through its online social circle, where marketing agents share messages with their personal audiences. Through sharing, liking, and commenting behaviors, messages can reach individuals exponentially. Compared to traditional methods that rely on offline materials, digital marketing leverages peer-to-peer communication patterns that reduce marketing costs (Duhon, 2015; Pulizzi, 2014; as cited in Hollebeek & Macky, 2019) and further enhance levels of building trust and engagement (Hollebeek & Macky, 2019). These characteristics suggest that marketing effectiveness in social media environments is closely tied to how messages diffuse through networks and how well they resonate with specific audiences.

Although social media marketing is widely adopted and overcomes many limitations of traditional approaches, such as high cost and low engagement, a persistent challenge remains: the misalignment between marketing messages and their intended audiences. Such misalignment often leads to weak engagement and limited diffusion performance. Therefore, investigating the determinants of marketing message propagation on social networks is critical, as improving

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content–audience alignment may substantially enhance marketing efficiency in the digital environment.

Previous studies identify several factors that can affect the effectiveness of message diffusion, which can be roughly classified into two categories: message content and social network. Emotional appeal, novelty, and relevance features of articles can affect users' desire to share. The network structure provides information related to the diffusion process (i.e. how the messages spread from one connected user to another, and users' behaviors). Although these factors have been extensively studied, a research gap remains regarding their interaction. Specifically, the performance of a marketing message depends not solely on the single content or social network, but on how the message fits within the social network for the audience.

When message characteristics align with audience preferences, engagement and sharing likelihood increase; when misalignment occurs, attention and participation decline. Despite this, some marketing systems still adopt a one-size-fits-all strategy that distributes identical content to all users without personalization. Such an approach underutilizes users' limited attention and sharing capacity and may weaken long-term network effectiveness.

1.2 Detailed description of the studied problem

While social media marketing has become a dominant communication channel, firms still face substantial uncertainty in predicting how marketing messages diffuse through social networks. Although both message content and network structure are known to influence information propagation, existing studies often examine them separately. This fragmented perspective limits

the ability to understand how content characteristics and network agent attributes jointly shape diffusion outcomes.

In practice, companies disseminate marketing content through employee-based or community-based networks, where diffusion performance depends not only on what is communicated (content features), but also on who shares it and how the network is structured (agent network characteristics). However, it remains unclear:

- How content features interact with network characteristics.
- Whether alignment (or misalignment) between message content and marketing agent attributes enhances diffusion efficiency.
- And how such alignment can be quantitatively modeled to predict marketing performance.

To clarify the previously ambiguous issues, this study centers on the following core research problem: How does alignment between content features and social network, specifically, marketing agents' network characteristics, influence the performance of marketing message diffusion in social media networks?

Based on the identified problem, the major objectives of this study are:

1. To investigate how message content features and diffusion network characteristics individually affect marketing performance.
2. To examine whether content–network alignment enhances diffusion outcomes by systematically comparing the three models: content-only, network-only, and aggregated

content-agent models, and to establish an integrated predictive model based on these to forecast diffusion performance.

3. To provide contributive ideas for designing optimized and cost-effective social marketing systems to improve marketing efficiency.

The key research issues:

Content Analysis:

- Examine content factors (topic distribution, article length, average reading time) that may affect diffusion outcomes.

Network Structure Analysis:

- Investigate diffusion network indicators (first and second layer width, maximum depth, re-share percentage, centrality, structural virality, Wiener index) that may relate to diffusion outcomes.
- Determine which network features truly represent diffusion structure rather than post-hoc statistical artifacts?

Content–Network Fit:

- Construct new interaction or combination variables from content and network analysis that may affect diffusion outcomes.
- Find connections between content-network personal matching, diffusion efficiency, and social capacity.

Data and Methods:

- Utilize diffusion records from the WeiMai system to build empirical measures of message spread.
- Determine the appropriate dependent variable for representing marketing performance.

- Apply statistical and machine-learning methods to relate message and network features to observe diffusion outcomes.

The following research tasks include:

Data Preparation: Extract and clean data from the WeiMai marketing system, construct diffusion-ready datasets with cascade linkage information.

Feature Engineering:

- Generate content features using LLM (Large Language Model)-based clustering and semantic representation.
- Compute diffusion network metrics (width, depth, structural virality, centrality, homophily). Organize data into three structured levels (the content, marketing agent, and marketing agent–topic levels).
- Construct interaction variables to capture network–content alignment. Distinguish structural diffusion indicators from non-diffusion attributes.

Descriptive and Exploratory Analysis: Summarize distributional patterns of content and network features. Examine preliminary correlations between features and diffusion performance. Identify potential alignment patterns.

Model Development and Comparison: Build baseline regression and advanced learning models to evaluate the determinants of diffusion outcomes. Compare different versions (specifications) of predictive models and test interaction effects to capture the features of content–network alignment.

Model Evaluation and Interpretation: Assess model accuracy using statistical accuracy metrics (R^2 , RMSE (Root Mean Square Error), etc.) and interpretability (feature importance, SHAP

(SHapley Additive exPlanations) analysis, etc.) to determine the most effective predictive hyper-parameter configuration.

Result Interpretation and Managerial Implications: Interpret how alignment influences diffusion efficiency. Translate findings into actionable recommendations for personalized message dissemination.

Research Scope:

The primary goal is to investigate how the content and network structure of personal matching affect the performance of diffusion processes in social media marketing, by analyzing a WeChat marketing dataset of a single interior design company, with a focus on feature engineering, quantitative modeling, and model interpretation. New skills to be acquired include using LLMs for topic modeling and analyzing data with new metrics.

Expected outcomes include:

- Three structured diffusion datasets integrating content, marketing agents' network, and marketing agent–topic level features.
- Empirical evidence identifying key content drivers of diffusion, structural network drivers, and the incremental contribution of network-content alignment.
- A predictive model for estimating diffusion outcomes under varying network conditions.
- Actionable ideas for data-driven and personalized social marketing strategies that move beyond uniform broadcasting.

1.3 Research Methodology

The proposed research is grounded in the WeiMai database platform, which operates a marketing system that serves customers (business) who need marketing, providing real-world data on the sharing and reading behavior (e.g., reading times, dwell time, and resharing activity) of articles from marketing agents, as well as the audience in the field of interior design. The data will be analyzed to examine how message features interact with network structures to measure reading volume and diffusion depth, and to construct a regression model to support the design of a marketing recommendation system for improved personal matching. The results of this research will help identify the key mechanisms that contribute to efficient marketing in a network, which can aid in designing personalized marketing strategies that break the barriers of the one-size-fits-all method.

The research methodology proceeds in three stages. First, human-resources records are matched with corresponding social media identifiers to integrate marketing agent attributes with online diffusion behavior. Second, a multi-modal large language model (GPT-4o) pipeline is deployed to process unstructured marketing articles, generating structured topic clusters and content features that enable content-based analysis. Third, diffusion cascades are reconstructed using sharing linkage records to correct layer assignments and parent mappings, forming a refined cascade network from which structural diffusion indicators are computed. Based on this reconstructed network, three interrelated datasets are developed at the content level, agent network level, and agent–topic level.

For the modeling stage, which is currently under development, regression methods will be employed as supervised learning approaches, where diffusion performance will serve as the dependent variable and content, network, and alignment features will serve as predictors. Model

performance will be evaluated using statistical measures such as R^2 and RMSE, and feature contribution analyses will be conducted to enhance interpretability.

2. Literature Review

In this section, existing studies involving social media marketing, message content, social network analysis, and personal matching are reviewed, and the research gaps that this study aims to fill are identified.

Beyond its conceptual importance, the effectiveness of social media marketing has been widely documented in both academic and industry research. Empirical evidence shows that online adults aged 18–34 are highly likely to follow brands on social networking platforms, with reported engagement rates reaching as high as 95% (LaMontagne, 2015). Social networks have also become the dominant source of inspiration for consumer purchasing decisions (PricewaterhouseCoopers Limited [PwC], 2018). From the perspective of practitioners, approximately 74% of marketers report that their social media marketing efforts are “somewhat effective” or “very effective” for their business outcomes (Read, 2018). Moreover, user-generated content plays a critical role in influencing consumer behavior, with more than half of consumers indicating that exposure to such content on social media increases their likelihood of purchasing a product (Curalate, 2017). Collectively, this evidence underscores the central role of social media platforms in influencing marketing effectiveness and consumer decision-making and highlights the importance of understanding how marketing messages diffuse through online social networks.

Content is essential in encouraging rebroadcasting activity (Berger & Schwartz, 2011). Berger and Schwartz (2011) investigated the influence of psychological factors on immediate and ongoing word of mouth, which is closely linked with diffusion and sales of the product. They analyzed the daily conversation data across multiple products and conducted both field and laboratory experiments. In detail, the results show that more interesting marketing messages can elicit additional immediate word of mouth from customers, and the content frequently **cued with the environment (triggered by environmental cues)** receives more ongoing word of mouth.

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Another research study by Heath et al. (2001) explored the emotional selection in **memes** that focused on the disgusting emotions in **urban legend stories**. Three experiments were conducted to control for informational factors and the extent of disgust in the legend story and coding analysis.

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Heath et al. (2001) found that the more disgust motifs contained in legend stories, the more widely distributed they are on online urban legend discussion forum, since they have the ability to evoke emotions of disgust. Finally, this research emphasized that emotional content was more likely to be shared (Heath et al., 2001). The social media engagement research by Lahuerta-Otero et al. (2018) identified the factors of the message content that enable tweet diffusion **and facilitate** electronic word of mouth (eWOM), by collecting and analyzing over thirty thousand tweets on Twitter. The multiple linear regression and theory of heuristic-systematic processing of information are implemented and illustrate factors, including tag, link, at, emotion, and length of tweet, that affect the dissemination and popularity (Lahuerta-Otero et al., 2018). These studies consistently demonstrate that message content characteristics significantly influence diffusion outcomes in online social networks.

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Katona et al. (2011) examined the network effect and personal influence in the diffusion of online social networks based on user growth data from the first 3.5 years of a major European social networking site that requires membership by invitation only. A binary choice model was developed to observe the impact of the network, influencer, and adopter on an individual's adoption decision (Katona et al., 2011). The outcome reveals two marked effects: the degree effect, representing that individuals are more likely to adopt if they are connected to many adopters, and the clustering effect, suggesting that the density of connections in the adopters' group positively influences the adoption tendency of individuals connected to this group (Katona et al., 2011). Both effects provide a valuable understanding of the impact of local network structure and influencers on the diffusion of information. Furthermore, the study of Yoganarasimhan (2012) also proposed a similar idea to that of Katona et al., but from analyzing a different social platform. Their study addressed the impact of social network structure on video propagation on YouTube, noting that the popularity of videos is closely related to the author's local network structure (Yoganarasimhan, 2012). Duffett (2015) investigated how advertising on Facebook affects people's purchase intentions among Millennials in South Africa by employing quantitative research and questionnaires, and analyzing the data using a generalized linear model. Research has shown that utilizing social networking to maintain relationships with customers makes customers more loyal to brands and makes brands appear more (Duffett, 2015). Overall, online social network marketing could bring additional benefits for businesses, and the network structure and influencers may significantly drive the network diffusion performance of marketing messages.

Customers acquired through referral programs tend to exhibit greater profit margins and lower churn rates than those acquired through other methods (Van Den Bulte et al., 2018). In response

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to this phenomenon, Van Den Bulte et al. (2018) analyzed the data from a German bank's referral program. One of the explanations they provided is that better matching between target customers and products, as well as social enrichment, impacts customer retention, and these referred customers have better matches with firms (Van Den Bulte et al., 2018). Zhang et al. (2017) noticed that previous studies have focused on the effects of content and personal influence on the spread of social media content separately, while researchers have paid little attention to investigating both together. Their study concentrated on simulating the role of content and influence in diffusion on social media, with additional consideration of the fit between messages and users and evaluating the relative effect of each factor (Zhang et al., 2017). Zhang et al. (2017) employed an individual-level split hazard model to scrutinize how users from top business schools diffuse Twitter posts, and the outcome illustrated the effects of message content, social influence, and content-user fit on the action of diffusing. Therefore, personal matching also has a significant impact on spreading information and providing businesses with higher profits and improved customer retention.

While prior work has studied the effects of content and network on online social media marketing, relatively few studies have considered them jointly, particularly in the context of personal matching. Building on Zhang et al.'s (2017) research, which focused on matching message and historical content preferences of first-degree connections to predict individual rebroadcasting, this study extends the analysis to explore how alignment between message characteristics and the broader social network structure influences overall spreading effect.

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3. Research Methodology and Initial Results

Stage 1:

In Stage 1, we constructed a diffusion-ready behavioral dataset by systematically cleaning, filtering, and integrating raw WeChat reading logs with agent attribute tables to ensure structural consistency for subsequent modeling. First, we extracted the full interaction log and removed records with invalid demographic codes (e.g., undefined gender) to prevent bias in later gender-composition and assortativity metrics. The cleaned dataset preserved reader-level diffusion fields such as `reader_layer`, `network_parent`, and sharing indicators, enabling later cascade reconstruction. Second, we standardized agent identifiers and merged departmental/job-role information through inner joins to retain only agents with complete attribute linkage, forming the basis for future profession–content alignment analysis. Third, we restricted the dataset to articles with verified DOCX content paths to ensure compatibility with the LLM-based content clustering pipeline in Stage 2. After these procedures, the final dataset contained 209,895 cleaned reading records across 1,051 articles and 6,942 unique (article, agent) pairs, providing a structurally validated and linkage-consistent foundation for subsequent content modeling and diffusion network construction.

Stage 2:

In Stage 2, we developed an automated LLM-based content analytics pipeline to transform each article’s full DOCX file (title, text, and images) into structured, quantitative topic representations. For each document, the system first extracted the title, textual content, word count, and embedded images using a custom `extract_docx` function, then uploaded up to 30 images to a GitHub repository and synchronized them via a content delivery network (CDN) to ensure stable

and low-latency multimodal access by the GPT-4o model. The LLM was prompted to assign weighted probabilities across a predefined set of content clusters, generating both title-level and content-level cluster distributions; these were converted into aligned numerical vectors and compared using vector-based similarity calculations (e.g., cosine-style alignment across ordered cluster arrays) to quantify title–content consistency. To ensure scalability and robustness, we implemented batch processing with randomized inter-call intervals to avoid API throttling, automatic retry logic for failed GPT calls, intermediate progress saving every few records, and post-run merging/deduplication of partial outputs into consolidated Excel datasets. The pipeline ultimately processed over 200 articles per batch and generated a merged results file containing structured cluster scores and derived features for 1,051 articles, providing the quantitative content variables required for the content-level and marketing agent–topic-level modeling in subsequent diffusion analyses.

Stage 3:

In Stage 3, we reconstructed the diffusion dataset to obtain structurally consistent cascade networks and to correct raw-log inconsistencies before computing any network metrics. Specifically, using the `network_parent` and `reader_layer` fields, we rebuilt each article’s propagation tree while performing layer corrections to ensure a proper cascade root definition by explicitly reassigning agent self-reads to layer 0. After these structural adjustments, we generated article-level cascade metrics including first-layer width, second-layer width, maximum depth, cascade size, and reshare rate. We then computed graph-theoretic indicators such as degree centrality, average out-degree, structural virality (Wiener index–based), and gender assortativity directly from the corrected adjacency structures, and aggregated these into content-level, agent-level, and agent–topic-level datasets. This reconstruction and correction step ensured that all

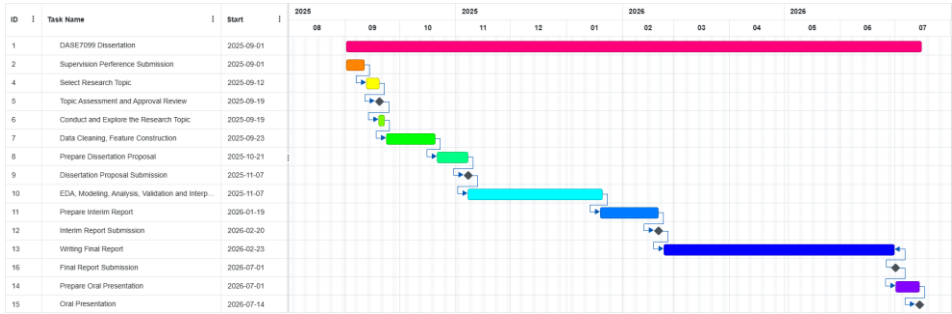
downstream diffusion measures were derived from valid cascades rather than from raw, which are potentially inconsistent interaction logs.

4. Progress review, major upgrade on research direction and approaches

Task	Sub-tasks	Completion	Action plan
A. Diffusion Data Engineering & Reconstruction	A1. Cleaning WeChat read/share logs and constructing cascade linkage fields	100%	
B. LLM-based Content Structuring Pipeline	B1. GPT-based topic clustering	100%	
	B2. Construction of agent–topic MatchScore & ProfessionContentMatch variables	100%	
C. Three-Level Dataset Construction	C1. Computation of cascade metrics (cascade_size, depth, first/second-layer width, reshare_pct, structural virality, centrality metrics)	100%	
	C2. Content-level, Agent-level, Agent–topic-level dataset	100%	
D. Modeling	D1. Content-only, Network-only, Combined model	0%	Implement baseline regression and ML models; Ensure predictors represent ex-ante network structure; exclude outcome-derived metrics.
E. Model Comparing and Integrating	E1. Model comparison and benchmarking	0%	Define evaluation metrics (RMSE, MAE, R ²); compare against native baseline.
	E2. Model integrating	0%	Building an integrated predictive model based on the performance of the three models

F. Result Analysis and Interpretation	F1. Result discussion, statistical interpretation, and theory linkage	0%	Strengthen interpretation of alignment mechanism; link results to homophily and diffusion literature without overstating causality.
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In the interim stage, the research issues have been refined to explicitly address two methodological challenges that were not clearly articulated before: (1) distinguishing diffusion-structure indicators from post-hoc statistical artifacts, and (2) determining an appropriate dependent variable to represent marketing performance while avoiding information leakage. These refinements strengthen the analytical validity and empirical defensibility of the study. Along with that, the research tasks have moved from general feature construction and modeling to a more structured workflow.



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